

Class 12 HW: Population Scale Analysis

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Population Scale Analysis

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes

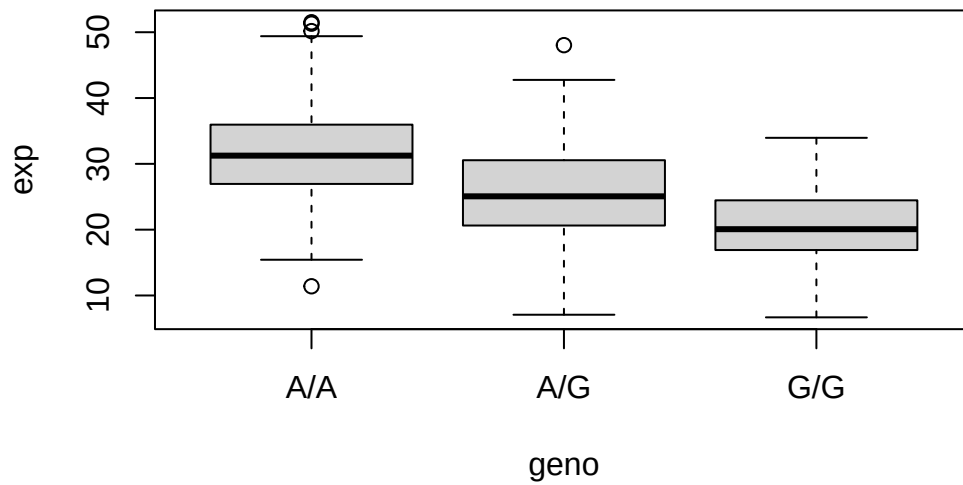
```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
table(expr$geno)
```

A/A	A/G	G/G
108	233	121

```
b <- boxplot(exp ~ geno, data = expr)
```



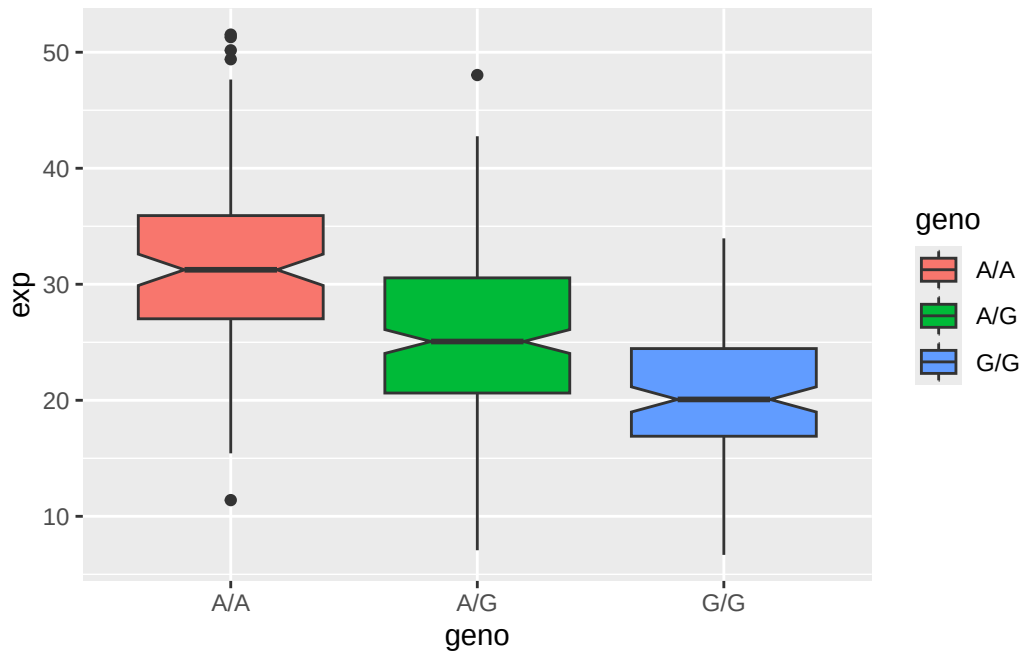
```
b$stats[3, ]
```

```
[1] 31.24847 25.06486 20.07363
```

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
```

```
ggplot(expr) + aes(geno, exp, fill=geno) +  
  geom_boxplot(notch=TRUE)
```



Based on the boxplot, ORM DL3 expression appears to differ by genotype. A/A individuals show the highest expression, G/G individuals show the lowest expression, and A/G individuals fall in between. This suggests rs8067378 may affect ORM DL3 expression, with the G allele associated with lower expression.